ModelIt!

Student Activity Pack

How Does 5G Actually Work?

Modeling Electromagnetic Wave Behavior in Wireless Communication Networks

Grade 10L2 | NGSS HS-PS4-2, HS-PS4-3 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think makes 5G different from 4G — and why is it faster?

2 Why do you think your phone signal gets worse inside certain buildings?

3 Draw a diagram showing how you think a signal travels from a cell tower to your phone.

4 What trade-offs do you think engineers have to make when designing a wireless network?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Signal Frequency | Wave Penetration | Data Transfer Rate | Tower Density | Interference Level | Energy Absorption | Network Capacity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: 4G Standard

Low frequency, wide coverage

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: 5G Millimeter Wave

High frequency, narrow coverage

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: 5G Mid-Band

Medium frequency, balanced approach

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Beamforming Precision — The ability of MIMO antenna arrays to focus electromagnetic energy into narrow beams directed at specific users, improving signal-to-noise ratio and allowing multiple users to share the same frequency without interference

Weather Impact — The effect of rain, snow, humidity, and atmospheric conditions on signal propagation — millimeter waves are particularly susceptible to rain fade, where water droplets absorb and scatter the signal

Latency — The time delay between transmitting and receiving a signal, measured in milliseconds — 5G promises 1ms latency versus 30-50ms for 4G, enabling real-time applications like autonomous vehicles and remote surgery

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does 5G promise lightning-fast internet but struggle to work through walls — and what trade-offs did engineers have to make? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the fundamental trade-off between data speed and signal penetration in electromagnetic communication?**

2. **Why does increasing Signal Frequency require such a dramatic increase in Tower Density — what does this reveal about wave physics?**

3. **What does your model suggest about why 5G networks use multiple frequency bands instead of one?**

4. **How does Interference Level change as Tower Density increases, and what engineering solutions address this challenge?**

5. **What are the limitations of modeling wireless network behavior with only seven components when real networks involve terrain, weather, and thousands of simultaneous users?**

8 STEM Challenge: Design a Campus 5G Network

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YOUR ENGINEERING MISSION

Design the placement and frequency strategy for a 5G network covering a school campus, optimizing for both coverage and capacity while managing interference.

THE SCENARIO

Your school district is deploying a 5G network across campus to support 2,000 simultaneous device connections for digital learning. The network must work inside classrooms, in outdoor spaces, and in the auditorium during events. Your team has been hired to design the antenna placement and frequency strategy.

Engineering Design Thinking

1. WHERE WOULD YOU PLACE SMALL CELLS FOR MILLIMETER WAVE COVERAGE AND MACRO CELLS FOR WIDE COVERAGE?
2. HOW WOULD YOU USE BEAMFORMING TO DIRECT SIGNALS TO CLASSROOM AREAS WITHOUT CREATING DEAD ZONES?
3. WHAT FREQUENCY MIX WOULD BALANCE SPEED, COVERAGE, AND COST FOR THE CAMPUS BUDGET?
4. HOW WOULD YOU TEST YOUR NETWORK DESIGN BEFORE INSTALLATION USING YOUR MODEL DATA?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

RF Engineers (Radio Frequency Engineers) design wireless communication systems, optimizing antenna placement, frequency selection, and interference management. They work for telecom companies, tech firms, and government agencies, earning $80,000–$150,000/year. Network Architects who design large-scale wireless infrastructure earn $100,000–$180,000/year.